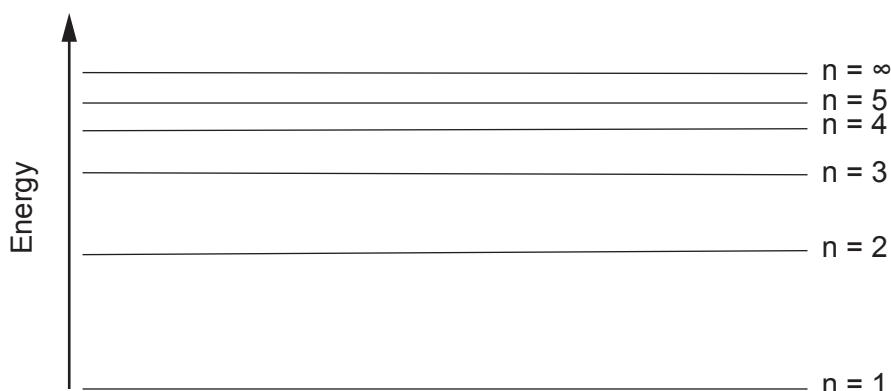


4. The diagram below shows the electron energy levels for a hydrogen atom.



- (a) On the diagram, draw an arrow to represent the transition corresponding to the ionisation of the atom. Label this arrow **A**. [1]
- (b) On the diagram, draw an arrow to represent the transition corresponding to the first line in the visible region in the atomic spectrum. Label this arrow **B**. [1]

5. A student said that $^{32}_{16}\text{S}^{2-}$ and $^{40}_{20}\text{Ca}^{2+}$ have the same electronic configuration.

Do you agree? Give a reason for your answer. [1]

.....

.....

6. Copper can be extracted from copper(II) oxide using hydrogen.

If the atom economy for this reaction is 78% and 4.2 g of copper is formed, calculate the total mass of the reactants. [2]

Mass of reactants = g

10

